

# ML Specification Report — playground-series-s3e3

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## Employee Attrition Prediction · our from-scratch agent (GBDT pool + tree-search v3)

*Figures grounded in the competition's `facts.json`, `eda_summary.json`, `STATUS.md`, `experiments_tree_v3.json`, and the leak-free benchmark log in `ext_facts.json`.*

## Overview

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The task is to rank 1,119 employees by attrition probability (metric: ROC-AUC). We solve it with a from-scratch pipeline — a pool of gradient-boosted trees (LightGBM / XGBoost / CatBoost) refined by four linear self-improvement rounds and a 22-node tree search that ends on a single heavily-regularized tuned-LGB champion. Our internal CV peaks at **OOF AUC 0.841442** (tree-search v3, node #11, 42 features). In the leak-free head-to-head benchmark our agent logs **AUC 0.8378**, while the NVIDIA reproduce-agent — copying a public ensembling kernel verbatim — reaches **0.8758**, so NVIDIA is ahead by ~0.038 AUC (~4.3%). This is a small, noisy 1,677-row dataset, so the honest headline is that the well-tuned public kernel retains a clear edge here.

**Why it matters.** Predicting which employees will leave lets organizations act before costly turnover — recruitment, onboarding, and lost institutional knowledge — by targeting retention where it pays off.

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## Data

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**Purpose of Data.** Predict whether an employee leaves the company (attrition) — a binary-classification Playground Series episode derived from the classic IBM HR attrition table. **Data Format** is clean **tabular CSV**, a mix of numeric and categorical columns. **Data Volume** is **1,677 training rows / 1,119 test rows**, with 33 raw features (**25 numeric + 8 categorical**) once `id` and the `Attrition` target are removed.

**Data Quality** is high: **no missing values** in either split, **0 duplicate rows**, and **no unseen test categories** in any categorical column. Two modelling wrinkles dominate: (a) strong class imbalance — only **11.9% of employees churned** (`Attrition` =1; 200 of 1,677), which forces stratified CV; and (b) three raw columns are **zero-variance constants** — `EmployeeCount` (=1), `StandardHours` (=80), `Over18` (=“Y”) — and are dropped. The strongest EDA signals, all consistent with HR intuition, were:

- **OverTime** is the single strongest categorical driver — **22.0% attrition when “Yes” vs 8.8% when “No”**.
- **MaritalStatus**: **Single 19.8% vs Divorced 4.9%** attrition.
- The strongest numeric correlations are all **negative (protective)**: `StockOptionLevel` (Spearman -0.227), `MonthlyIncome` (-0.184), `YearsAtCompany` (-0.170), `Age` (-0.169), `JobLevel` (-0.163), `TotalWorkingYears` (-0.160).
- Heavy internal collinearity: tenure features (`YearsAtCompany` / `YearsInCurrentRole` / `YearsWithCurrManager`) correlate 0.75–0.79, and `JobLevel` ↔ `MonthlyIncome`  $r \approx 0.91$  — a

redundancy the tree search later exploits.

- **No single-feature leakage** (no column reaches AUC > 0.75 alone). With only 1,677 rows, **per-fold AUC swings 0.79–0.89** — expected variance, not a bug.

**Annotation Guidelines.** The label is `Attrition`  $\in \{0, 1\}$  (1 = left the company). Submissions are **attrition probabilities** scored by AUC — a ranking, not a hard 0/1 decision.

**Feature Set.** From the 33 raw columns we drop 3 constants and engineer up to **48 features**, grouped below:

| Group                       | Features                                                                                                                                                                                                                                                           |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encoded (label + frequency) | 7 categoricals — <code>BusinessTravel</code> , <code>Department</code> , <code>EducationField</code> , <code>Gender</code> , <code>JobRole</code> , <code>MaritalStatus</code> , <code>OverTime</code> (each $\rightarrow$ <code>_le</code> + <code>_freq</code> ) |
| Tenure ratios               | <code>role_tenure_ratio</code> , <code>mgr_tenure_ratio</code> , <code>promo_ratio</code> , <code>company_tenure_ratio</code>                                                                                                                                      |
| Income                      | <code>income_per_joblevel</code> , <code>income_per_year_worked</code>                                                                                                                                                                                             |
| Satisfaction                | <code>satisfaction_avg</code> , <code>satisfaction_min</code>                                                                                                                                                                                                      |
| Interaction / derived       | <code>overtime_joblevel</code> , <code>age_at_join</code> , <code>companies_per_year</code>                                                                                                                                                                        |
| Dropped (zero variance)     | <code>EmployeeCount</code> , <code>StandardHours</code> , <code>Over18</code>                                                                                                                                                                                      |

Encoding is **label + frequency**, not WOE/target encoding — our pipeline deliberately avoids the weight-of-evidence route (which belongs to the NVIDIA benchmark; see the Evaluation section). The champion node trims further to **42 features**, dropping the 5 raw tenure columns plus `income_per_year_worked` once their engineered ratios exist.

**Splitting strategy.** A single canonical **Stratified 5-Fold CV on `Attrition`** (`shuffle = True`, `seed = 42`), held **fixed across every experiment and every tree-search node** so scores are directly comparable. Given the tiny 1,677-row split, a full 5-fold pass is cheap enough to run inside each Optuna trial (no fold-0 proxy needed). The leaderboard was not used (unattended run — submission files generated but not uploaded), so **all scores here are CV-only OOF AUC**.

## Models & Architecture

**Purpose of Architecture.** A binary attrition classifier that maximizes ROC-AUC. **Architecture Type** is a **gradient-boosted decision-tree pool** — LightGBM as the primary learner, with XGBoost and CatBoost members available for blending. Unusually for this project, the CV **champion is a solo tuned-LGB model** (tree-search node #11), which beat every weighted / rank-average blend the search tried on this small dataset.

Each member takes the same **Input Format** — a numeric feature matrix with categoricals label/frequency-encoded — of **Input Dimension 42 features** for the champion (48 in the full pool), a 1-D vector per row with no spatial/sequence structure.

**Architecture Description.** Each GBDT member is an additive ensemble of decision trees grown by histogram-based, leaf-wise gradient boosting. The champion is deliberately **shallow and heavily regularized** to fight overfitting on 1,677 rows: LightGBM with `num_leaves = 3`, `max_depth = 4`, `min_child_samples = 60`, `n_estimators` up to 2000 with early stopping, `learning_rate  $\approx$  0.0503`.

Where blends are used, the layer on top is a convex weighted average (Dirichlet-sampled) or a rank-average of member OOF predictions. **Model Complexity** is best expressed as trees  $\times$  leaves rather than a dense parameter count: with only 3 leaves per tree, each LGB member is an extremely low-capacity learner by design — the search repeatedly confirmed that *regularization over capacity* is what this dataset rewards.

## Training procedures

Training proceeded as **four linear self-improvement rounds followed by a tree search**, all on the shared 5-fold split and measured on the same OOF scale:

| Stage | Configuration                                               | OOF AUC         |
|-------|-------------------------------------------------------------|-----------------|
| 1     | generic LGB+XGB+CAT blend (no feature engineering)          | 0.81624         |
| 2     | + 48 engineered features, imbalance-weighted 3-model blend  | 0.819008        |
| 3     | imbalance weighting removed, lighter/regularized LGB        | 0.832925        |
| 4     | Optuna-tuned LGB (50-trial full-CV-AUC objective, solo)     | 0.837305        |
| 5     | + seed-bag, 5-way weighted blend                            | 0.837776        |
| 6     | CatBoost native categoricals (diagnostic, solo)             | 0.814259        |
| 7     | 6-way rank-average blend (linear-iteration best)            | 0.838140        |
| 8     | tree-search v3 (22 nodes; node #11 = tuned-LGB on 42 feats) | <b>0.841442</b> |

The key Reflexion finding at stage 3: because **AUC is a ranking metric**, `scale_pos_weight` / `class_weights` mostly perturbed the loss landscape rather than helping ranking — *removing* all imbalance weighting jumped LGB from 0.819  $\rightarrow$  0.833. The champion (stage 8, node #11) traces to a comp-local hypothesis — dropping the redundant raw tenure columns once their ratios exist — verified as a real +0.0041 gain over the tuned-LGB root.

The **Loss Function** is binary log-loss (LightGBM `binary` objective) with model selection on AUC. The **Optimization Algorithm** is gradient boosting (histogram, leaf-wise) — **not** SGD/ADAM — with blend weights found by Dirichlet sampling / weight search. The **Learning Rate** is the boosting shrinkage (champion LGB `learning_rate`  $\approx$  0.0503 ; the LGB\_orig member used 0.03); there is **no Learning Rate Scheduler** (N/A — *GBDT convergence is governed by CV early-stopping, not a schedule*) and **no Batch Size** (N/A — *full-dataset histogram boosting, not mini-batched*).

**Training Duration** for the tree-search v3 sweep was  **$\sim$ 34 s of wall-clock over 22 model fits** (22 evaluated nodes, 0 failed); the one-time Optuna LGB tune was 50 trials / 111.3 s, and a solo LGB fits in  $\sim$ 1 s. **Training Memory Consumption Limits** were not explicitly capped — a  $1,677 \times 48$  float matrix is trivial on the single arm64 CPU machine; peak memory was **not recorded**. **Transfer Learning** is N/A (*no pretrained weights*); its analogue here is **experience-library prior injection** — `suggest_priors` returned 7 bullets (P0–P6) that shaped the mutation queues (directionally correct: 0 wasted searches down closed dead-ends like re-tuning CatBoost, though the biggest score-mover was a comp-local idea, not a transferred one). **Data Augmentation Techniques** are N/A (*tabular*); the analogues are feature engineering (the 48 features) and

seed-bagging (a seed = 2024 bag, which the weight search zeroed out — little extra diversity on an already-directly-optimized model).

**Reproducibility Standards** are strict: a fixed fold seed (42) shared across all experiments, logged Optuna and seed-bag seeds, and a faithful-replay contract. Per the benchmark log, a faithful replay of the committed tree reproduces **22/22 nodes exactly ( $\Delta = 0$ )**.

## Inference procedures

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**Decision Threshold** is *N/A* — AUC is a ranking metric, so we submit raw probabilities and never threshold (a 0.5 cut would only be needed if hard labels were required). **Inference Duration** was **not separately profiled**; GBDT scoring of the 1,119 test rows is sub-second per member on CPU. **Inference Memory Consumption Limits** are negligible and **not capped** — tree inference on a ~1k-row test set is lightweight.

## Evaluation & Benchmarking

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**Task for Performance Evaluation.** Rank the test employees by attrition probability. **Performance Metrics.** ROC-AUC is the sole competition metric; our internal CV climbs **0.81624** → **0.819008** → **0.832925** → **0.837305** → **0.837776** → **0.838140** → **0.841442** across the eight stages, with the tree-search v3 champion (node #11) topping out at **OOF AUC 0.841442**. In the leak-free benchmark harness our agent logs **AUC 0.8378**. The competition leaderboard was not used (unattended run), so every figure here is **CV / OOF, not public-LB**.

**Performance Benchmarking.** The natural comparator is the **NVIDIA reproduce-agent**, which copies the strongest public kernel verbatim:

| Agent     | Approach                                                            | AUC           | Note           |
|-----------|---------------------------------------------------------------------|---------------|----------------|
| NVIDIA    | reproduces <code>chunweishen/ps-s03e03-ensembling</code> (verbatim) | <b>0.8758</b> | winner         |
| Our agent | from-scratch GBDT pool + tree-search v3                             | 0.8378        | -0.038 (~4.3%) |

NVIDIA leads by ~0.038 AUC (~4.3%). The relevant caveat concerns **weight-of-evidence (WOE) encoding**, which lives in the **NVIDIA kernel, not our pipeline** (ours used label/frequency encoding throughout). When that kernel is audited for target leakage, its WOE turns out to be **fold-safe** — de-leaking leaves the score **unchanged at 0.8758**, so NVIDIA’s win here is **genuine**, not a leakage artifact. On this small, noisy ~1.7k-row S3-era problem the well-optimized public ensembling kernel holds a clear, honest edge over our from-scratch pipeline; the reproduce-vs-originate gap narrows only on the larger, newer seasons analyzed separately.

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## Final-Architecture Re-Run (v5 pipeline, Stage 0.5 in-lane) — 2026-08

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All 20 baseline competitions are being re-run in an isolated root ( `~/benchruns/myagent-rerun` ) under one frozen architecture ( `docs/rerun_manifest.json` ): every lane runs the **full v5 pipeline including the Stage 0.5 problem dossier** (dossier.json written in-lane before modeling), holds no credentials, and is checked by a

per-lane transcript audit. Leaderboard scores for the re-run are **TBD** until the separate credentialed operator scoring step; the figures below are the lane's own local CV only and revise nothing in the sections above.

- **Lane:** done 2026-08-14, 20 min.
- re-run local CV: **ROC-AUC 0.841605** pooled 5-fold OOF, equal-weight blend of four seed-bagged members (no fitted weights, no optimism correction needed).
- Public / private leaderboard for this re-run: **TBD** (pending operator scoring).